

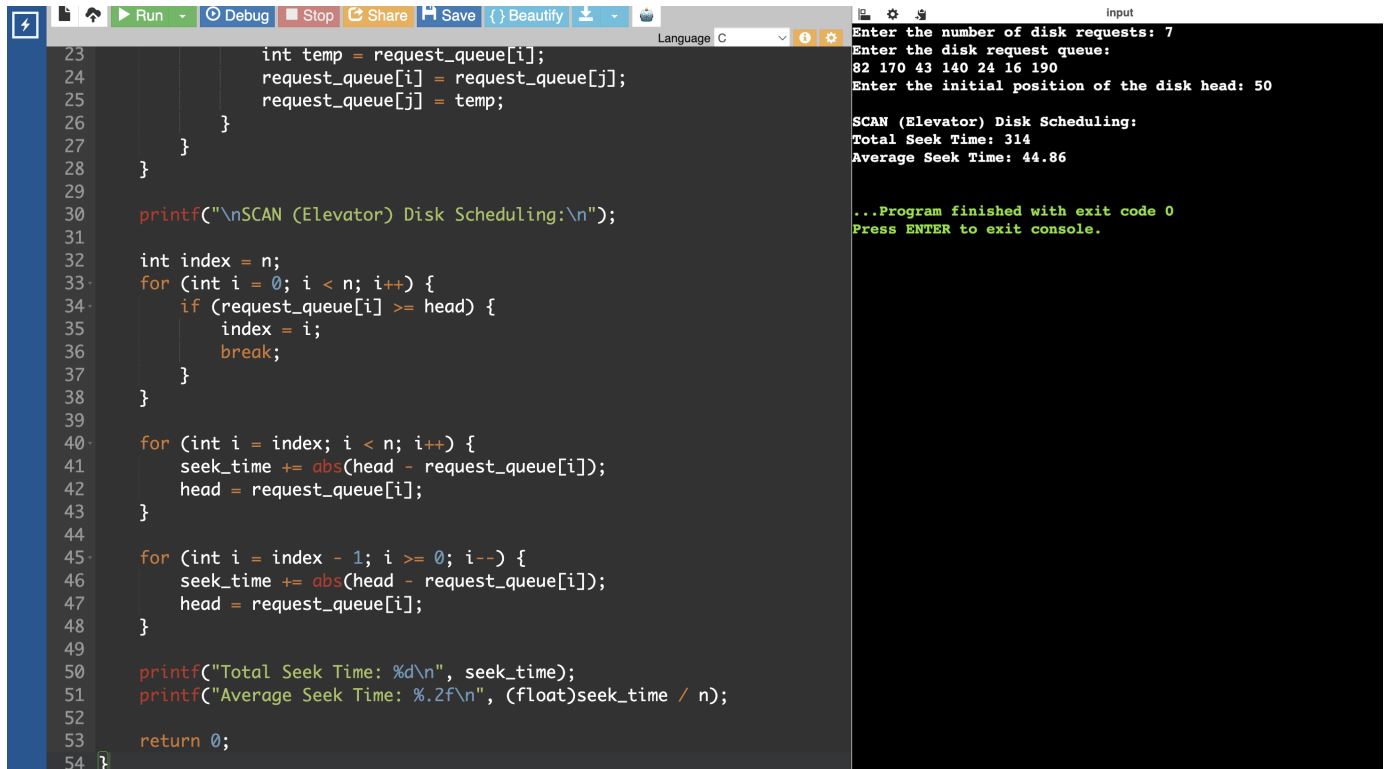
CSA0404 – OPERATING SYSTEMS

EXPERIMENT 38

QUESTION

Design a C program to simulate SCAN disk scheduling algorithm.

PROGRAM



```
23     int temp = request_queue[i];
24     request_queue[i] = request_queue[j];
25     request_queue[j] = temp;
26 }
27 }
28 }
29
30 printf("\nSCAN (Elevator) Disk Scheduling:\n");
31
32 int index = n;
33 for (int i = 0; i < n; i++) {
34     if (request_queue[i] >= head) {
35         index = i;
36         break;
37     }
38 }
39
40 for (int i = index; i < n; i++) {
41     seek_time += abs(head - request_queue[i]);
42     head = request_queue[i];
43 }
44
45 for (int i = index - 1; i >= 0; i--) {
46     seek_time += abs(head - request_queue[i]);
47     head = request_queue[i];
48 }
49
50 printf("Total Seek Time: %d\n", seek_time);
51 printf("Average Seek Time: %.2f\n", (float)seek_time / n);
52
53 return 0;
54 }
```

Input

Enter the number of disk requests: 7
Enter the disk request queue:
82 170 43 140 24 16 190
Enter the initial position of the disk head: 50

SCAN (Elevator) Disk Scheduling:
Total Seek Time: 314
Average Seek Time: 44.86

...Program finished with exit code 0
Press ENTER to exit console.